

PITTSFIELD, MA, USA

WMO: 744104

Lat: 42.427N

Lon: 73.289W

Elev: 1194

StdP: 14.07

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 14763

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.1	2.6	-13.1	2.8	1.1	-8.4	3.7	4.6	25.8	21.8	23.8	21.4	7.1	300	0.512

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	85.7	70.7	83.0	69.2	80.7	68.1	73.3	82.1	71.8	79.5	70.4	77.4	8.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.4	117.0	77.7	69.4	112.9	76.5	67.9	107.0	75.0	37.6	82.4	36.3	79.8	35.1	77.1	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.5	18.5	16.3	-9.3	89.9	4.5	2.1	-12.6	91.4	-15.2	92.7	-17.7	93.9	-21.0	95.4		
(5)			-9.5	76.2	4.7	1.5	-12.9	77.4	-15.7	78.3	-18.4	79.1	-21.8	80.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.6	22.2	24.4	32.5	45.4	56.3	63.9	69.4	67.5	60.7	49.1	38.2	28.6
	DBStd	18.40	11.36	10.10	9.91	8.66	7.80	6.14	4.83	5.33	6.84	8.22	8.61	9.47
	HDD50	3493	863	717	549	189	25	1	0	0	2	118	365	664
	HDD65	7084	1327	1136	1008	590	289	91	15	35	166	495	804	1128
	CDD50	2266	0	1	6	51	220	419	600	543	322	91	11	2
	CDD65	379	0	0	0	2	18	59	150	113	35	2	0	0
	CDH74	3062	0	0	0	51	241	527	1185	787	260	11	0	0
	CDH80	654	0	0	0	10	45	117	279	152	50	1	0	0
Wind	WSAvg	6.0	7.4	7.7	7.7	7.2	5.5	4.8	4.4	4.1	4.3	5.7	6.3	7.1
Precipitation	PrecAvg	45.0	3.1	2.5	3.3	3.1	3.7	4.4	4.6	4.4	4.4	4.4	3.3	3.8
	PrecMax	59.6	7.0	7.1	6.2	6.3	7.2	8.2	11.3	10.1	8.9	13.9	6.3	7.2
	PrecMin	34.1	1.3	1.0	0.8	1.3	0.9	1.4	2.1	2.1	1.4	1.1	0.2	1.7
	PrecStd	6.8	1.4	1.2	1.3	1.3	1.7	2.1	2.3	2.2	2.2	2.7	1.3	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.0	58.9	68.0	80.8	84.5	87.7	89.8	88.1	85.3	74.9	65.5	59.4
		MCWB	53.7	52.0	53.9	58.7	66.3	72.1	73.7	72.9	71.6	63.3	57.6	54.5
	2%	DB	48.3	50.3	58.0	72.7	80.5	83.5	85.8	83.9	80.7	70.5	61.2	52.4
		MCWB	43.5	43.1	48.1	55.8	65.1	69.4	71.1	70.9	68.1	62.8	55.0	49.1
	5%	DB	41.3	43.9	52.2	66.7	76.1	79.7	83.1	81.1	76.6	66.8	56.9	47.2
		MCWB	37.9	38.1	44.5	53.3	62.0	66.9	69.7	69.2	66.3	60.5	51.6	44.0
	10%	DB	36.9	38.2	47.2	61.4	71.7	76.6	80.5	78.7	73.1	63.0	52.8	42.3
		MCWB	34.0	34.3	41.6	49.5	59.7	64.9	68.6	67.7	64.6	56.8	47.8	39.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.0	53.9	55.5	63.2	69.9	74.0	75.3	75.2	73.1	67.5	60.3	56.1
		MCDB	55.5	58.4	62.9	71.9	80.3	84.6	85.6	83.6	80.7	71.8	64.2	58.6
	2%	WB	44.4	44.2	50.6	59.3	67.5	71.5	73.4	72.9	70.6	64.9	56.3	49.8
		MCDB	47.0	49.2	57.2	70.3	77.2	80.0	82.7	80.5	77.5	68.6	59.7	52.4
	5%	WB	38.4	38.7	45.9	55.4	64.9	69.5	71.9	71.2	68.4	61.5	52.4	44.3
		MCDB	41.1	42.9	50.0	64.5	72.5	76.8	79.7	78.2	74.3	66.3	56.0	46.7
	10%	WB	34.2	34.6	41.4	51.4	62.4	67.3	70.5	69.6	66.2	57.3	48.3	39.2
		MCDB	36.6	38.1	46.8	58.6	69.6	73.9	77.6	75.5	71.5	61.6	52.2	42.0
Mean Daily Temperature Range	5% DB	MDBR	15.0	16.5	17.0	20.3	20.6	19.2	19.5	19.5	20.2	17.1	16.2	13.3
		MCDBR	19.6	23.0	24.4	29.9	27.3	24.0	22.4	21.4	23.8	22.3	22.3	19.5
		MCWBR	17.5	17.9	16.0	16.0	13.7	11.8	10.4	10.2	12.8	14.4	16.8	17.9
	5% WB	MCDBR	19.2	21.9	22.1	26.2	23.4	20.4	19.1	18.3	19.5	19.2	19.5	18.4
		MCWBR	17.9	18.2	17.1	16.1	13.1	11.2	9.9	9.5	11.6	14.2	16.9	17.6
Clear-Sky Solar Irradiance	taub	0.286	0.303	0.322	0.342	0.378	0.422	0.441	0.415	0.357	0.336	0.303	0.284	
	taud	2.401	2.354	2.412	2.430	2.360	2.256	2.213	2.271	2.418	2.491	2.536	2.499	
	Ebn at Noon	273	285	293	294	284	271	264	267	278	271	266	266	
	Edh at Noon	25	30	33	35	38	43	44	40	32	27	22	21	
All-Sky Solar Radiation	RadAvg	520	801	1132	1444	1637	1774	1849	1645	1341	849	582	418	
	RadStd	41	79	92	92	157	173	161	121	88	113	89	58	48

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (71 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+187	+95	

Nomenclature: See separate page